Error Semitransparent Universal Control of a Bosonic Logical Qubit

Error transparency and pulse optimisation

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Presenting Roy, Wetherbee & Fatemi, arXiv:2603.15356
with our simulation replication

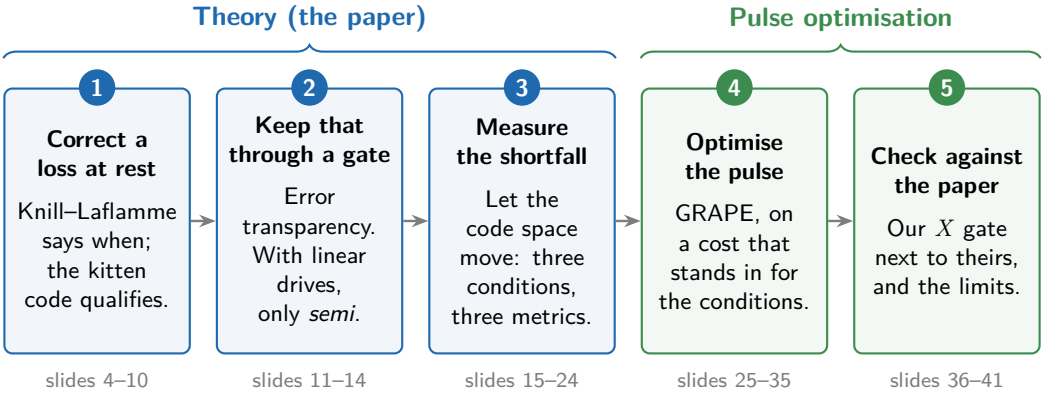
October 26, 2026

Why gates are the hard part

- **Memories work.** Bosonic codes in a cavity now beat break-even while *idling*: the encoded qubit outlives the best uncorrected one.
- **Gates are harder.** A photon lost *during* a gate is pushed around by the rest of the gate, and can end up uncorrectable.
- **Error-transparent gates** fix this in principle.
- **This paper:** ordinary linear drives plus *moving* code spaces give X, H, T gates on a binomial code that are *almost* transparent.

Break-even with bosonic codes: e.g. Ni et al., Nature 616 (2023); Sivak et al., Nature 616 (2023).

Where we are going



Exact transparency is not expected. The paper's move is to make the shortfall something we can *measure* (3) and *optimise* (4).

From step 4 on, [ours] marks where our replication differs.

The device: a cavity dispersively coupled to a transmon

$\hat{a}$: cavity (storage oscillator, holds the logical qubit). $\hat{q}$: transmon (ancilla, for control and parity readout). In the dispersive regime:

$$H(t) = H_a + H_q + H_{\text{int}} + H_{\text{drive}}(t) \qquad \text{[Eq. A1]}$$

$$H_a/\hbar = \omega_a\, \hat{a}^\dagger \hat{a} + \tfrac{K}{2}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K'}{6}\, \hat{a}^{\dagger 3}\hat{a}^3 \qquad \text{[A2]}$$

$$H_q/\hbar = \omega_q\, \hat{q}^\dagger \hat{q} + \tfrac{K_q}{2}\, \hat{q}^{\dagger 2}\hat{q}^2 \qquad \text{[A3]}$$

$$H_{\text{int}}/\hbar = \chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q} + \tfrac{\chi'}{2}\, \hat{q}^\dagger \hat{q}\, \hat{a}^{\dagger 2}\hat{a}^2 \qquad \text{[A4]}$$

$$H_{\text{drive}}(t)/\hbar = \epsilon(t)\, \hat{a} + \Omega(t)\, \hat{q} + \text{h.c.} \qquad \text{[A5]}$$

$\chi/2\pi$	-3.66 MHz	$K/2\pi$	-22 kHz	$K_q/2\pi$	-180 MHz
$\chi'/2\pi$	39 kHz	$K'/2\pi$	590 Hz	cavity T_1	180 μs

Main-text Eq. 9 is the same Hamiltonian at low order: it drops K' and χ'. Values from Table I.

Reading the Hamiltonian

- **Dispersive χ.** $(\omega_q + \chi\, \hat{a}^\dagger \hat{a})\, \hat{q}^\dagger \hat{q}$: the transmon frequency moves by χ per photon, so a transmon drive is *photon-number selective*.
- **Kerr K, K', χ'.** Small photon-number-dependent shifts. K and χ' (and, weakly, K') are what break error transparency (Eq. A6).
- **Rotating frame.** Removes ω_a, ω_q; the static part left over (χ, Kerr, K_q) is the H_0 our simulation uses.
- **Controls.** Only the complex $\epsilon(t)$ and $\Omega(t)$, which GRAPE shapes.
- **The error the code targets.** Cavity photon loss: jump operator $\hat{a}$, rate $\kappa = 1/T_1$.

Notation clash

$\epsilon(t)$ is a *drive amplitude*. The noise *channel* on the next slides is written $\varepsilon(\rho)$, a different object.

Setup: code space, projector, errors

A logical qubit lives in a **code space** $\mathcal{C} \subset \mathcal{H}$, with orthonormal codewords $\{|c_i\rangle\}$ and projector

$$P_{\mathcal{C}} = \sum_i |c_i\rangle\langle c_i| = UU^\dagger, \qquad P_{\mathcal{C}}^\dagger = P_{\mathcal{C}}, \quad P_{\mathcal{C}}^2 = P_{\mathcal{C}},$$

where U is the encoding isometry ($U^\dagger U = \mathbb{I}_{\text{logical}}$).

Noise is a channel in **Kraus form**:

$$\varepsilon(\rho) = \sum_j E_j \rho E_j^\dagger, \qquad \sum_j E_j^\dagger E_j = \mathbb{I}.$$

The Kraus set is not unique

Any unitary recombination $F_k = \sum_j W_{kj} E_j$ gives the *same* channel. We will use this freedom.

The Knill–Laflamme theorem

Statement

A recovery operation correcting $\{E_a\}$ exists **if and only if** there is a Hermitian matrix $\alpha = (\alpha_{ab})$, $\alpha_{ab} \in \mathbb{C}$, with

$$P_{\mathcal{C}}\, E_a^\dagger E_b\, P_{\mathcal{C}} = \alpha_{ab}\, P_{\mathcal{C}} \qquad \forall a, b.$$

Equivalently, in codeword form:

$$\left\langle i_L \middle| E_a^\dagger E_b \middle| j_L \right\rangle = \alpha_{ab}\, \delta_{ij}.$$

Two independent demands hide in that one equation:

- δ_{ij}: errors must not **mix** codewords with each other.
- α_{ab} *independent of* i: the environment must learn nothing about **which** codeword it hit.

Why that condition is enough (sketch)

1. Diagonalise α and recombine the errors to match (same channel). Then

$$P_{\mathcal{C}}\, F_k^\dagger F_l\, P_{\mathcal{C}} = d_k\, \delta_{kl}\, P_{\mathcal{C}}.$$

2. So each error moves the code **rigidly** onto its own copy $\mathcal{C}_k$, and different copies are **orthogonal**.
3. **Recovery:** measure which copy (the syndrome), then rotate it back. Outcome k has probability d_k for every logical state.

In plain words: errors scatter the code into orthogonal, undistorted copies of itself. Find the copy, rotate it back. The measurement learns the error, never the logical state.

Full derivation: backup slides B1–B3.

The binomial “kitten” code

$$|0_L\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}}, \qquad |1_L\rangle = |2\rangle$$

Both codewords live on the **even-parity** Fock states.

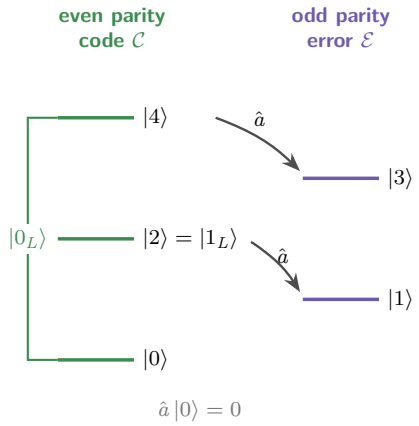
A single photon loss $\hat{a}$ sends them to

$$\hat{a}\,|0_L\rangle = \sqrt{2}\,|3\rangle \equiv \sqrt{2}\,|0_E\rangle\,,$$

$$\hat{a}\,|1_L\rangle = \sqrt{2}\,|1\rangle \equiv \sqrt{2}\,|1_E\rangle\,,$$

entirely on the **odd-parity** states.

[Sec. III]



Why the kitten code satisfies Knill–Laflamme

Two facts, and they are exactly the two demands from the theorem:

1. **Distinct, orthogonal images.** $|0_L\rangle \to |0_E\rangle$, $|1_L\rangle \to |1_E\rangle$ with $\langle 0_E|1_E\rangle = \langle 3|1\rangle = 0$. Loss acts as an **isometry** onto the error space, so codewords do not mix.
2. **Equal coefficients.** Both images carry the same $\sqrt{2}$, i.e.

$$\left\langle 0_L \middle| \hat{a}^\dagger \hat{a} \middle| 0_L \right\rangle = \left\langle 1_L \middle| \hat{a}^\dagger \hat{a} \middle| 1_L \right\rangle = 2.$$

   Equal mean photon number $\Rightarrow$ no which-codeword information.

Together:

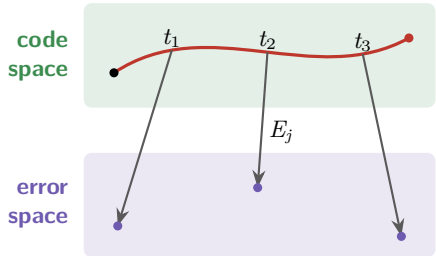
$$P_{\mathcal{C}}\, \hat{a}^\dagger \hat{a}\, P_{\mathcal{C}} = 2\, P_{\mathcal{C}} \qquad \text{(proportional to the logical identity).}$$

In plain words: the environment learns *that* a photon left, and nothing about *which* state it left from. No which-path information, so no logical dephasing (to first order in κt).

Why a gate breaks all of that

Idling. The codewords barely move (only small Kerr phases). A lost photon lands on nearly the same known error word, parity detects it, and you recover.

During a gate. The drive moves the state, so mid-gate it is no longer a codeword.



Photon lost at t: $\underbrace{U(T,t)}_{\text{finish}}\underbrace{E_j}_{\text{loss}}\underbrace{U(t,0)}_{\text{start}}|\psi_L(0)\rangle$ depends on t, and t is **random**.

In plain words: a photon lost mid-gate ends up somewhere different for every loss time. The recovery at the end has one fixed target, so it cannot undo them all.

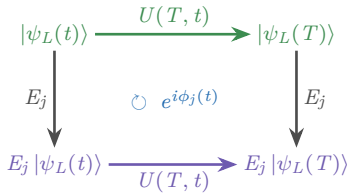
Error transparency: the jump commutes through the gate

The demand: for all $t \in [0, T]$,

$$U(T,t)\, E_j\, U(t,0)\, |\psi_L(0)\rangle = e^{i\phi_j(t)}\, E_j\, U(T,0)\, |\psi_L(0)\rangle \qquad \text{[Sec. II]}$$

- **Left:** evolve to t, get hit by E_j, then finish the gate.
- **Right:** finish the whole ideal gate, then get hit, times a scalar phase.

If it holds $\forall t$, a photon lost at *any* time gives the **ideal gate output with one error applied at the end**: a single, known, correctable configuration. One recovery at $t = T$ cleans it up.



Both routes reach the same state, up to $e^{i\phi_j(t)}$, for *every* t.

When does a loss commute through? Error closure

Push the jump past the Hamiltonian. A loss at any time stays correctable if, at every instant, the commutator lands back among the correctable errors:

$$\big[\, H(t),\, \hat{a} \,\big] \;\in\; \mathrm{span}\{\text{correctable errors}\} \qquad [\text{App. A}]$$

This is *error closure*. Exact transparency, $[H, \hat{a}] = \lambda\, \hat{a}$, is the special case.

Term by term, for the Hamiltonian of slide 4:

Term	**Commutator with $\hat{a}$**	**Correctable?**
$\omega_a\, \hat{a}^\dagger \hat{a}$, $\chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q}$	$\propto \hat{a}$ (times $\hat{q}^\dagger \hat{q}$)	yes
linear drives $\epsilon\, \hat{a} + \Omega\, \hat{q} + \mathrm{h.c.}$	$-\epsilon^*$, a number	yes
Kerr K, higher dispersive χ'	$\propto \hat{a}^\dagger \hat{a}^2 = \hat{n}\, \hat{a}$	no

The obstruction: nonlinear terms that will not commute

The system Hamiltonian (slide 4) carries the cavity self-Kerr $\frac{K}{2}\hat{a}^{\dagger 2}\hat{a}^2$ and a higher-order dispersive term $\frac{\chi'}{2}\hat{a}^{\dagger 2}\hat{a}^2\hat{q}^\dagger\hat{q}$. Their commutator with the jump operator is

$$\left[\ \frac{K}{2}\hat{a}^{\dagger 2}\hat{a}^2 + \frac{\chi'}{2}\hat{a}^{\dagger 2}\hat{a}^2\,\hat{q}^\dagger\hat{q},\ \ \hat{a}\ \right] = -\left(K + \chi'\,\hat{q}^\dagger\hat{q}\right)\hat{a}^\dagger\hat{a}^2. \qquad \text{[Eq. A6]}$$

- $\hat{a}^\dagger\hat{a}^2 = \hat{n}\,\hat{a}$ is a jump weighted by photon number. It hits $|0_L\rangle$ and $|1_L\rangle$ with different strengths (3 vs 1), so it violates the "α independent of i" half of Knill–Laflamme.
- A linear drive can soften its effect along a trajectory, but cannot cancel it: its own commutator with $\hat{a}$ is just a number, with no $\hat{n}$-dependent part (App. A).

Therefore: error *semi*transparent

Exact ET is not expected with linear drives alone. Accept a small, nonzero violation in exchange for simple, standard control, and then **measure how much you recover**.

Letting the code space move

The conditions for *static* ET ask the state to stay inside the fixed $\mathcal{C}(0)$ and $\mathcal{E}(0)$. Gate drives sweep it far outside both. So: **track moving subspaces** instead.

Instantaneous code space [Eq. 1]

$$\mathcal{C}(t) = \left\{ U(t,0)\,|\psi_L(0)\rangle \;:\; |\psi_L(0)\rangle \in \mathcal{C}(0) \right\}, \qquad P_{\mathcal{C}}(t) = U(t,0)\,P_{\mathcal{C}}\,U^{\dagger}(t,0).$$

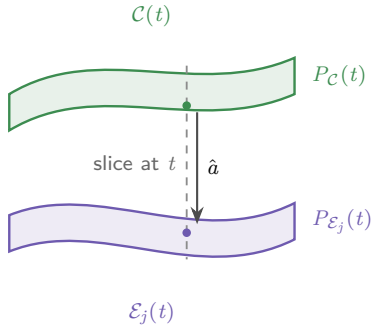
- A 2-dimensional subspace carried along the **ideal trajectory**.
- $U(t,0)$ is the *ideal, lossless* propagator of the driven system.

And the error space with it

Instantaneous error space [Eq. 2]

$$\mathcal{E}_j(t) = \mathrm{span}\Big\{\, E_j\,|\psi_L(t)\rangle \;:\; |\psi_L(t)\rangle \in \mathcal{C}(t) \,\Big\}$$

$$P_{\mathcal{E}_j}(t) \;\propto\; E_j\, P_{\mathcal{C}}(t)\, E_j^\dagger$$



For photon loss, $E_j = \hat{a}$:

$$\mathcal{E}(t) = \mathrm{span}\Big\{\hat{a}\,U(t,0)\,|0_L\rangle\,,\;\hat{a}\,U(t,0)\,|1_L\rangle\Big\}$$

In plain words: where a photon loss *right now* would send you, given the ideal trajectory.

The three conditions

The dynamic commuting square holds if **three things are true at every instant**:

1. Correctable

The moving code space is still a valid QEC code.

Knill–Laflamme, instant by instant

2. Tracking

The error space goes where the drive carries it.

evolved space = instantaneous space

3. Matched

Code and error spaces feel the same internal dynamics.

same generator, up to a phase rate

Each one gets an equation, and later a number that measures how badly it is violated.

Condition 1: instantaneous correctability

Over the error set $E = \{\mathbb{I},\ \hat{a}\}$, every projected error product must be **proportional to the logical identity**:

$$P_{\mathcal{C}}(t)\, E_i^{\dagger} E_j\, P_{\mathcal{C}}(t) = c_{ij}(t)\, P_{\mathcal{C}}(t), \qquad \forall i, j, t. \qquad \text{[Eq. 3]}$$

This is Knill–Laflamme again, now demanded of the code space *as it moves*.

The kitten code at $t = 0$:

$$c(0) = \begin{pmatrix} c_{II} & c_{Ia} \\ c_{aI} & c_{aa} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

At $t = 0$ the only live entry is c_{aa}, and $\hat{a}^{\dagger}\hat{a} = \hat{n}$, so it is the demand

$$\langle 0_L|\hat{n}|0_L\rangle = \langle 1_L|\hat{n}|1_L\rangle \ \ (= 2).$$

Condition 2: subspace tracking

$$U(t,0)\,\mathcal{E}_j(0) = \mathcal{E}_j(t), \qquad \forall j, t. \qquad \text{[Eq. 4]}$$

Two different objects that must coincide:

- **Evolved** $U(t,0)\mathcal{E}_j(0)$: where a jump at $t = 0$ has been *carried to* by now.
- **Instantaneous** $\mathcal{E}_j(t)$: where a jump *right now* would land.

Why anchoring at $t = 0$ is enough

A jump can happen at any time s. Right after it the state is in $\mathcal{E}_j(s)$, and the drive carries it to $U(T,s)\,\mathcal{E}_j(s)$ by the end T.

$$U(T,s)\,\mathcal{E}_j(s) \underset{\text{Eq. 4 at } s}{=} U(T,s)\;U(s,0)\,\mathcal{E}_j(0) \underset{\text{compose}}{=} U(T,0)\,\mathcal{E}_j(0) \underset{\text{Eq. 4 at } T}{=} \mathcal{E}_j(T).$$

So an error j always ends in the same subspace $\mathcal{E}_j(T)$, whenever it happened.

Condition 3: matched internal dynamics

With $P_j(t) \propto E_j P_{\mathcal{C}}(t)$ the map carrying the code space onto the instantaneous error space (an isometry up to normalisation),

$$P_j^\dagger(t)\, H(t)\, P_j(t) = P_{\mathcal{C}}^\dagger(t)\, H(t)\, P_{\mathcal{C}}(t) + d_j(t)\, P_{\mathcal{C}}(t), \qquad \forall j, t, \qquad \text{[Eq. 5]}$$

where $d_j(t)$ is a number. The paper calls it complex; both sides are Hermitian, so it is real.

Read as Hamiltonians restricted to the two subspaces:

$$H_{\mathcal{E}}(t) = H_{\mathcal{C}}(t) + d_j(t)\, \mathbb{I}.$$

- The two subspaces are driven by the **same generator**, differing only by a multiple of the identity.
- A shift proportional to $\mathbb{I}$ is a global phase rate, physically invisible.

In plain words: the drive must push the error space around in exactly the same way it pushes the code space, apart from an overall phase.

Three metrics, one per condition

Each condition gets a scalar that is **zero when the condition holds**.

Condition	Metric	Question it answers
1	QEC violation $\Delta_{\mathrm{QEC}}(t)$ (Eq. 6)	Is the moving code still correctable?
2	Leakage $L_{\mathcal{E}_j}(t)$ (Eq. 7)	Is the error state in the right subspace?
3	Trajectory mismatch $\eta_{\mathcal{E}_j,\psi}(t)$ (Eq. 8)	Is it in the right place inside it?

All three are **time-dependent**: they are evaluated at every step of the pulse, not just at $t = T$.

Metric 1: QEC violation $\Delta_{\mathrm{QEC}}(t)$

Form the 2×2 projected error matrices, over $E = \{\mathbb{I}, \hat{a}\}$:

$$M_{ik}(t) = P_{\mathcal{C}}(t)\, E_i^\dagger E_k\, P_{\mathcal{C}}(t) = c_{ik} P_{\mathcal{C}} + x_{ik} X_{\mathcal{C}} + y_{ik}\, Y_{\mathcal{C}} + z_{ik} Z_{\mathcal{C}}. \qquad \text{[Eqs. A8–A9]}$$

The $P_{\mathcal{C}}$ part is correctable; the X, Y, Z parts are **not**. So strip the identity and measure what is left:

$$\Delta_{\mathrm{QEC}}(t) = \sum_{i,k} \Big\| M_{ik} - \tfrac{1}{2} \operatorname{Tr}[M_{ik}]\, P_{\mathcal{C}} \Big\|_2 = \sum_{i,k} 2\big(|x_{ik}|^2 + |y_{ik}|^2 + |z_{ik}|^2\big). \qquad \text{[Eqs. 6, A11]}$$

In plain words: $\Delta_{\mathrm{QEC}}(t) = 0$ means a photon lost at time t can still be corrected. Larger means a bigger uncorrectable part. It is a size, not a probability.

$\|A\|_2 = \operatorname{Tr}(A^\dagger A)$ is the squared Frobenius norm (App. A): no square root, hence the factor of 2.

Metric 2: instantaneous leakage $L_{\mathcal{E}_j}(t)$

$$L_{\mathcal{E}_j}(t) = 1 - \mathrm{Tr}\big[\, P_{\mathcal{E}_j}(t)\, \rho(t) \,\big] \qquad \text{[Eq. 7]}$$

where $\rho(0)$ is the **maximally mixed state in the static error space** $\mathcal{E}_j(0)$, evolved unitarily with no losses:

$$\rho(0) = \frac{P_{\mathcal{E}_j}(0)}{d} = \tfrac{1}{2}\big(|0_E\rangle\langle 0_E| + |1_E\rangle\langle 1_E|\big), \qquad \rho(t) = U(t,0)\, \rho(0)\, U^\dagger(t,0).$$

So the trace compares two subspaces: the **evolved** error space (carried in $\rho(t)$) against the **instantaneous** one (in $P_{\mathcal{E}_j}(t)$). That is exactly Condition 2.

In plain words: the probability, averaged over all logical inputs, that a state which suffered the error at $t = 0$ is found *outside* the subspace an error at time t would produce.

Metric 3: trajectory mismatch $\eta_{\mathcal{E}_j,\psi}(t)$

$$\eta_{\mathcal{E}_j,\psi}(t) = \left\| \vec{r}_{\mathcal{C}}(t) - \vec{r}_{\mathcal{E}_j}(t) \right\|_2 \in [0,2] \qquad \text{[Eq. 8]}$$

$$\vec{r}_{\mathcal{C}}(t) = \mathrm{Tr}\left[\vec{\sigma}_{\mathcal{C}}(t)\,\rho_{|\psi\rangle}(t)\right], \qquad \vec{r}_{\mathcal{E}_j}(t) = \mathrm{Tr}\left[\vec{\sigma}_{\mathcal{E}_j}(t)\,\tilde{\rho}_{\left|\psi_{E_j}\right\rangle}(t)\right]$$

- Take a logical state $|\psi\rangle$ and its error partner $\left|\psi_{E_j}\right\rangle \propto E_j\,|\psi\rangle$; evolve both.
- Write each as a Bloch vector in **its own** instantaneous logical frame ($\vec{\sigma}_{\mathcal{C}}(t)$ in the code space, $\vec{\sigma}_{\mathcal{E}_j} \propto E_j \vec{\sigma}_{\mathcal{C}} E_j^\dagger$ in the error space), and measure the distance.
- The error state is *projected into* $\mathcal{E}_j(t)$ first, so $\eta_{\mathcal{E}_j,\psi}$ is blind to leakage: independent information from $L_{\mathcal{E}_j}$.
- Evaluated over all 6 Bloch cardinal states.

In plain words: "where is the state on the code sphere?" vs. "where is it on the error sphere?" Zero means no relative rotation (Condition 3).

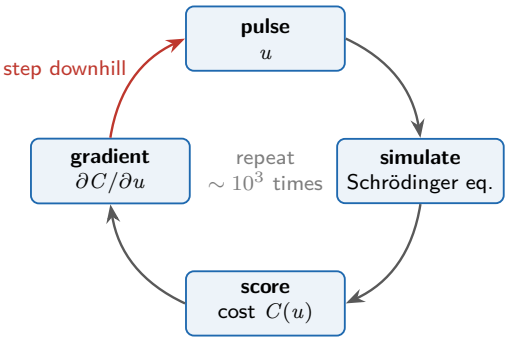
What GRAPE does (Gradient Ascent Pulse Engineering)

The pulse as a list of numbers.

Cut the pulse into N steps of $\Delta t = 1$ ns, constant on each step. Each step holds 4 real numbers: I and Q of the cavity and transmon drives (slide 27). Stack them end to end:

$$u = \big(u(t_1),\, u(t_2),\, \ldots,\, u(t_N)\big),$$

a single vector of $4N$ numbers. [App. C]



In plain words: GRAPE is guided trial and error. Simulate the pulse, score how far the test states land from their targets, and let the gradient say which way to nudge each of the $4N$ numbers. All are nudged together. It is a local search: it usually converges in about a thousand rounds, but where it lands depends on the random start.

The model the optimiser simulates

Go to the frame rotating with both modes, drives on resonance. Then ω_a, ω_q drop out, and what is left of slide 4 is the static **drift**:

$$H_0/\hbar = \chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q} + \tfrac{K}{2}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K'}{6}\, \hat{a}^{\dagger 3}\hat{a}^3 + \tfrac{\chi'}{2}\, \hat{q}^\dagger \hat{q}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K_q}{2}\, \hat{q}^{\dagger 2}\hat{q}^2.$$

[from Eqs. A2–A4]

cavity	Fock $\lvert 0\rangle, \ldots, \lvert 19\rangle$ $(n_c = 20)$	pulses reach $\sim \lvert 10\rangle$; stable for 16–28
transmon	$\lvert g\rangle, \lvert e\rangle, \lvert f\rangle$	keeps leakage to $\lvert f\rangle$
total	dimension 60	[our truncation]

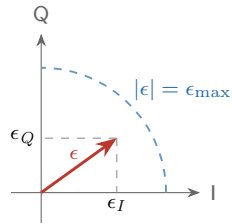
Lossless and unitary

Dissipation is not simulated: the evolution is purely unitary. Photon loss appears only inside the cost, as the operator $\hat{a}$.

Rewriting the drive as real controls: I and Q

Split each complex drive (Eq. 10) into two real parts, $\epsilon = \epsilon_I + i\,\epsilon_Q$ and $\Omega = \Omega_I + i\,\Omega_Q$:

$$\epsilon\,\hat{a} + \epsilon^*\,\hat{a}^\dagger = \epsilon_I\,(\hat{a} + \hat{a}^\dagger) + \epsilon_Q\; i(\hat{a} - \hat{a}^\dagger).$$



Same for the transmon. So

$$H(t) = H_0 + \sum_{j=1}^{4} u_j(t)\,H_j,$$

$$u = (\epsilon_I,\ \epsilon_Q,\ \Omega_I,\ \Omega_Q),$$
$$H_j = \big(\hat{a} + \hat{a}^\dagger,\ i(\hat{a} - \hat{a}^\dagger),\ \hat{q} + \hat{q}^\dagger,\ i(\hat{q} - \hat{q}^\dagger)\big).$$

In plain words: the optimiser only sees real numbers, and H is linear in them. I and Q are also exactly the two channels of the IQ mixer that makes the pulse in the lab.

Piecewise-constant pulses and the gradient

Hold u fixed on each step of length Δt. On step k:

$$H_k = H_0 + \sum_j u_j(k)\, H_j, \qquad U_k = e^{-iH_k \Delta t}, \qquad U(T,0) = U_N \cdots U_2\, U_1.$$



- The pulse is constant on each step, so each step is one matrix exponential.
- The whole gate is these N matrices multiplied in order.
- Each number in u affects only its own step, so one sweep forward and one sweep back through the product give the gradient for all $4N$ numbers at once.

What goes in, and what should come out

Inputs. The six cardinal states of the logical Bloch sphere, transmon in $|g\rangle$,

$$|\pm Z_L\rangle\,, \quad |\pm X_L\rangle\,, \quad |\pm Y_L\rangle\,,$$

and their error partners $|\psi_E\rangle \propto \hat{a}\,|\psi_C\rangle$ in the error space (ψ_C: a code state, the ψ_L of Part I).

Targets. The ideal logical gate applied to each input, with the transmon back in $|g\rangle$.

Gates and time grid

gate	length	drives	steps N
X	1 μs	cavity + transmon	1000
H	1 μs	cavity + transmon	1000
T	0.6 μs	transmon only	600

Piecewise constant, $\Delta t = 1$ ns. [App. C]

In plain words: every gate is tested on six states at once. A pulse that sends all six to the right place has the right logical action, relative phases included.

The cost function: one weighted sum

$$C_{\text{tot}} = \sum_i w_i \, C_i \qquad \text{[Eq. C8]}$$

Term	Question it asks	Evaluated
C_1 gate infidelity	Did the gate happen?	end of pulse
C_2 transparency	Does a loss *now* land where an earlier one does?	every step
C_3 velocity variance	Is the motion spread over the gate?	every step
C_4 amplitude cap	Is the drive within 4 MHz?	every step [ours]

Eqs. 6–8 are diagnostics, not the objective

The optimiser uses **state overlaps**, which are easy to differentiate; the metrics only check the result. Not one cost per condition: C_2 pushes on Conditions 2 and 3 together, nothing targets Condition 1 directly, and C_3 is a regulariser.

C_1: did the gate happen?

$$C_1 = 1 - F_{\text{target}} = 1 - \frac{1}{6}\sum_{i=1}^{6} \left| \langle \psi_{\text{final},i} | \psi_{\text{target},i} \rangle \right|^2 \qquad \text{[Eq. C1]}$$

- Evolve each of the six cardinal states through the **whole** pulse.
- Compare it with where the ideal gate sends it. Only $t = T$ matters.
- The cost is averaged over all six cardinal states, so it also checks the relative phase between $|0_L\rangle$ and $|1_L\rangle$, not just the populations.
- This is **all** an ordinary ("Ord") gate is optimised for.

In plain words: C_1 evaluates where the logical states end up at the end of the gate, not how they got there. A photon lost halfway through is invisible to it.

C_2: transparency at every step

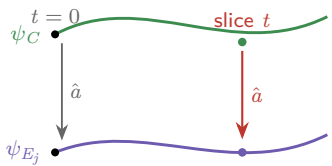
$$F_{\mathrm{ET}} = \frac{1}{N}\sum_{i=0}^{N-1} \frac{\left|\langle\psi_{E_j}(t_i)|\hat{a}|\psi_C(t_i)\rangle\right|^2}{\mathcal{N}}, \qquad \mathcal{N} = \sqrt{\langle\psi_C(t_i)|\hat{a}^\dagger\hat{a}|\psi_C(t_i)\rangle} \qquad \text{[Eq. C2]}$$

Average over the six cardinal pairs: $C_2 = 1 - F_{\mathrm{ET,avg}}$. [Eq. C3]

Evolve ψ_C and ψ_{E_j} **separately**, both from $t = 0$. At each step:

- $\hat{a}\,|\psi_C(t)\rangle$: where a loss *now* would send you.
- $|\psi_{E_j}(t)\rangle$: where a loss at $t = 0$ has been carried by now.

Overlap $= 1$ means the two agree.



We divide by $\mathcal{N}^2$, not $\mathcal{N}$, so that $F_{\mathrm{ET}} \leq 1$ (Cauchy–Schwarz). See notes.

C_3: keep the dynamics spread out

The loophole. A state that barely moves is nearly transparent for free. So the optimiser can idle for most of the pulse and do the whole gate in a short burst at the end. C_2 averages over every step, so it reads high, although the burst is not transparent.

The fix. Measure the speed of the state through Hilbert space, and penalise *uneven* speed:

$$v_i = \frac{2}{\Delta t} \arccos \left| \langle \psi_C(t_i) | \psi_C(t_{i+1}) \rangle \right| \qquad \text{[Eqs. C4–C5]}$$

$$\sigma_j^2 = \frac{1}{N} \sum_{i=0}^{N-1} \left(v_i - \bar{v} \right)^2, \qquad C_3 = \frac{1}{6} \sum_{j=1}^{6} \sigma_j^2 \qquad \text{[Eqs. C6–C7]}$$

- $2 \arccos |\langle \cdot | \cdot \rangle|$ is the distance between neighbouring states, so v is a speed.
- A **regulariser**: it does not improve transparency by itself.

We use $\sigma_j^2 / \bar{v}^2$, which has no units. See notes.

Putting it together: two stages

Stage	(w_1, w_2, w_3)	Goal	What happens
1	$(1,\ 0.6\text{–}0.75,\ 5\text{–}10)$	build transparency	F_{ET} up to 0.83, fidelity 0.9–0.99
2	$(1,\ 0.1,\ 0)$	finish the gate	transparency changes by $\sim 1\%$

[App. C]

The control experiment

Ord = the same optimiser with C_1 alone. EsT and Ord differ only in the cost, so a transparency gap can be pinned on the cost, but only once it is larger than the spread between random starts.

Hardware constraints on the pulse

Constraint	Value	Purpose
Bandwidth	± 50 MHz about the carrier	matches the AWG; Fourier-cut the pulse
Amplitude	$\lvert\epsilon\rvert/2\pi,\ \lvert\Omega\rvert/2\pi \leq 4$ MHz	maximum drive strength
Ramp	48 ns Gaussian rise and fall	start and end at zero drive
T gate	cavity drive off	transmon only

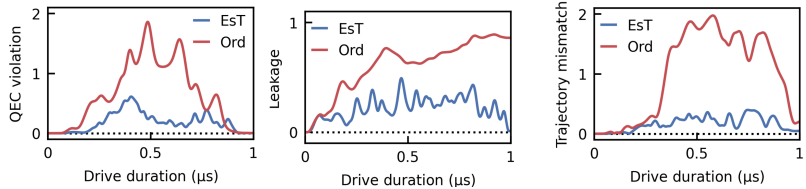
[App. C]

How they are imposed

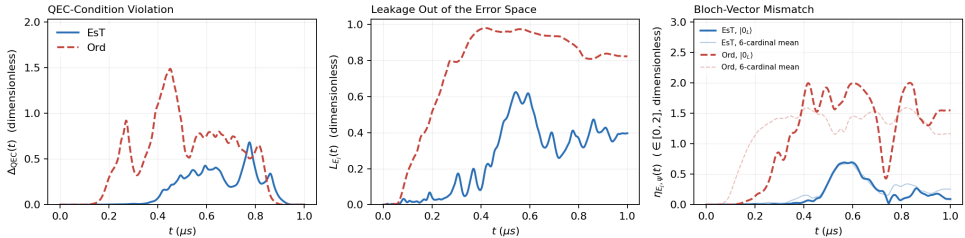
In our code the shape constraints are built into the pulse: $x \to$ band-limit $\to$ ramp $\to$ mask $\to u$. The amplitude cap is a *circle* in the I–Q plane (slide 27), because the amplitude of the drive is the magnitude of the vector (I, Q), i.e. $\sqrt{I^2 + Q^2}$. [ours]

Our X gate vs the paper: the three metrics in time

Paper, Fig. 1d–f



Ours, Fig. 2a



Left to right: Δ_{QEC}, L, η for $|0_L\rangle$ (ours also shows the six-cardinal mean, faint). X gate, $1\,\mu$s, lossless simulation, linear axes. EsT blue, Ord red.

What agrees, and what does not

X gate, EsT / Ord	Paper	Ours
L, time-averaged	0.22 / 0.61 digitised	0.258 / 0.740
Ord : EsT on L	2.74	2.87
$L(T)$, end of gate	0.013 / 0.862	**0.396** / 0.824
Δ_{QEC}, peak	≈0.6 / ≈1.9	0.68 / 1.49
$\eta(\lvert 0_L\rangle)$ at $t = T$	≈0.05 / ≈0.2	0.09 / **1.55**

≈: read by eye from the paper's figure.

Agrees

- EsT below Ord through mid-pulse, on all three.
- Leakage level and Ord:EsT ratio within about 20%.

Differs

- The end of the gate: their EsT finishes transparent, ours still leaks.
- Our late Δ_{QEC} peak (0.78 μs).

Why the endpoint differs, and caveats

The end-of-gate leak

- **Not the printed cost form.** Retraining with the paper's own C_2 and C_3 leaves $L(T)$ at about 0.4.
- **A terminal error-space term fixes it.** Adding error-space fidelity at $t = T$: $L(T)$ $0.40 \to 0.005$ at $F_1 = 0.9995$ (one seed).
- **Hypothesis:** the paper's EsT gate "optimizes for fidelity of operation in both code and error space" (App. C), but no such term is printed.

Caveats

One seed per variant. Fock levels are not matched (EsT 9, Ord 8; the paper matches them). Paper values are digitised or read by eye.

What this approach does not cover

- **Only cavity photon loss is made transparent.** Transmon decay and dephasing during the gate are not. With $T_{2,R}^{q} = 30\ \mu$s against a $1\ \mu$s gate, they are comparable to cavity loss ($\bar{n}\ T / T_1 \approx 1$–$2\%$).
- **Single loss only.** Two photons lost in one gate are not covered.
- **The optimisation is lossless.** It never trades transparency against decoherence: nothing stops a pulse buying transparency with extra time in $|e\rangle$.
- **Model vs. hardware.** Line distortion and mixer imbalance are not modelled, and the $|f\rangle$ shift is taken as 2χ (from $\chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q}$).
- **Exact transparency is not reached**, by design (slide 14).

Summary

1. **QEC while idling**: Knill–Laflamme, $P_{\mathcal{C}} E_a^\dagger E_b P_{\mathcal{C}} \propto P_{\mathcal{C}}$. The kitten code meets it, and parity is the syndrome.
2. **Gates break it**: a loss at a random time gives a t-dependent endpoint. Error transparency asks the jump to commute through the gate.
3. **Exact ET is not expected** with linear drives: Kerr and χ' leave an uncorrectable $\hat{a}^\dagger \hat{a}^2$. Hence "semi".
4. **Moving subspaces** $\mathcal{C}(t)$, $\mathcal{E}_j(t)$: three conditions, and three metrics that measure them along the whole pulse.
5. **GRAPE** shapes a staircase of real I/Q controls on a lossless model of the device.
6. **The cost**: C_1 fidelity at the end, C_2 transparency at every step, C_3 to spread the dynamics over the gate, in two stages. Ord is C_1 alone.
7. **Constraints**: 50 MHz bandwidth, 4 MHz amplitude, 48 ns ramps, 1 ns steps.

The trade: give up exact transparency, keep the standard linear cQED toolkit, and optimise for as much transparency as it allows.

Our replication: status and open questions

Done (simulation only)

- Independent model and cost, with two gradient codes (autodiff and hand-derived) that agree.
- X, H, T trained, EsT and Ord, with two documented changes to the printed cost (C_2, C_3).
- On X and H, EsT lowers leakage and trajectory mismatch about 3–7$\times$ against Ord (one seed; slide 37).
- Extra costs tried ($\Delta\bar{n}$, end-of-gate error-space fidelity, smoothness): each moves its own target; none is a free win.

Open

- Seed-to-seed spread of the EsT–Ord gap.
- End-of-gate leakage: $L(T) \approx 0.4$ on X, against the paper's 0.013 (slide 38).
- Transmon errors are outside the model.
- A Fock-matched, multi-seed comparison with the paper.

Why that condition is enough (1/2): untangle the errors

Shorthand on these backup slides: $P \equiv P_{\mathcal{C}}$.

1. α is Hermitian, so a unitary u diagonalises it:

$$\alpha = u\, d\, u^{\dagger}, \qquad d = \mathrm{diag}(d_k), \quad d_k \geq 0.$$

2. Recombine the Kraus operators with it (same channel, slide 6): $F_k = \sum_a u_{ak}\, E_a$.

3. The error-correction condition becomes

$$P\, F_k^{\dagger} F_l\, P = (u^{\dagger} \alpha\, u)_{kl}\, P = d_k\, \delta_{kl}\, P.$$

Only the diagonal survives

Different errors $F_k \neq F_l$ no longer overlap on the code, and error k occurs with probability d_k whichever codeword it hits.

Why that condition is enough (2/2): undistorted copies

Normalise each error on the code:

$$V_k \equiv \frac{1}{\sqrt{d_k}} F_k P.$$

Each V_k is a **partial isometry**: it maps $\mathcal{C}$ onto an image $\mathcal{C}_k = V_k \mathcal{C}$ without distorting it, and

$$V_k^\dagger V_k = P, \qquad V_k^\dagger V_l = 0 \;\; (k \neq l).$$

So the errors scatter the code into **disjoint, orthogonal, undistorted copies** of itself.

Correcting a discrete set also corrects its entire linear span, so discrete errors suffice.

Recovery: find the copy, rotate it back

After the noise, the code sits on copy k with probability d_k: $\varepsilon(\rho) = \sum_k d_k \, V_k \, \rho \, V_k^\dagger$.

1. **Error-space projector** onto $\mathcal{C}_k = V_k \mathcal{C}$:

$$P_k = V_k \, V_k^\dagger, \qquad P_k P_l = \delta_{kl} \, P_k.$$

2. **Measure** $\{P_k\}$. Outcome k and the state it leaves:

$$p_k = \mathrm{Tr}\big[P_k \, \varepsilon(\rho)\big] = d_k, \qquad \varepsilon(\rho) \;\to\; V_k \, \rho \, V_k^\dagger.$$

3. **Rotate back** with $V_k^\dagger$:

$$V_k^\dagger \big(V_k \, \rho \, V_k^\dagger \big) \, V_k = P \rho P = \rho.$$

$p_k = d_k$ for every ρ: the syndrome reveals the error, never the logical state.